Improving Horizontal-Well Productivity Using Novel Technology and Optimization of Drilling Fluid

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Summary

Horizontal wells have dominated conventional wells because of their increased production rates, improved recovery efficiency, increased reservoir drainage area, and delayed gas and water coning. However, nonuniform-flow profiles can result in premature water/gas production, loss of production and reserves, and a decrease in profitability, which will shorten the life, and defeat the purpose, of a horizontal well.

This paper describes a new production-technology system that will optimize production, delay water/gas coning, eliminate/minimize annular flow, and ensure uniform inflow along the lateral at the cost of a minute pressure drop in long, high-rate horizontal wells. A case history is presented in which this production technology, combined with sand-control technology, has resulted in significant savings for the operator and improved production in offshore horizontal oil wells.

Introduction

A mature clastic reservoir was drilled offshore Saudi Arabia. The Z-field was put into production in 1972, with 220 wells drilled to date. The majority of these wells are vertical wells targeting the 3.5-darcy high-permeability massive sands in the principal reservoir of the field.

Recently, to increase oil production, the development and drilling strategies on an eight-well platform were switched to drill horizontal wells in all but one well—the platform locator (see **Fig. 1**). Six of the horizontal wells were completed with an average of 3,400 ft of cemented 7-in. liner. Only the best-quality pay (25 to 40% of the horizontal section) was perforated by use of costly 3⅜-in. tubing-conveyed-perforation (TCP) guns.

In addition to the rig time and high cost involved in this type of completion technique, high risk is also taken by having the TCP guns on rig location, because it raises a number of safety concerns.

A new production-completion system was being evaluated in the last horizontal well (Z-253) in the same platform. The evaluation results of this new production-completion system showed a minimum potential saving of U.S. $240,000, with three rig-days ahead of the drilling schedule, by eliminating running, cementing, cleaning, and perforating of the 7-in. liner. The decision was finally made to run the safer and more cost-effective 2,200-ft new production-completion system to meet the main objective of extending the life of the horizontal Z-253 well.

A Uniform Inflow-Control System

Flux profiles in horizontal offshore wells can be extremely nonuniform, particularly in fractured carbonate reservoirs. It was, therefore, decided to complete Z-253 using an inflow-control system to ensure uniform inflow along horizontal wells, regardless of permeability, formation damage, or location in the wellbore.

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The system incorporates a standalone premium screen fitted with a proprietary low-velocity flow regulator. For nonsanding reservoirs, uniform-flow modules are available without screens, which lowers overall system costs.

The inflow control device (ICD) consists of up to three helical-flow channels, as shown in **Fig. 2,** spinning the flow before entry into the wellbore. The helical channels can be modified to suit a variety of downhole-flow conditions. The objective of this ICD is to prevent highly productive zones from producing too much; hence, allowing the pressure in the wellbore to be lowered to pull harder on the less-productive zones. If the well path is closed to either water and/or gas, it will prevent any undesired coning effects that lead to premature water and/or gas breakthrough.

The system was deployed in the open hole of the horizontal well with an external casing packer (ECP). The number of ICDs and ECPs used was a function of well length and reservoir heterogeneity, as well as target delivery rates. There is no technical limit on the number of sections that can be placed, but recent cases have demonstrated a practical/commercial limit of 10 to 15 packers per well.

Zonal isolation in each 600 ft of lateral section **(Fig. 3)** demonstrated the ability to balance the inflow into the screen liner with the specially designed ICD. The ICD uses the helical channels, shown in Fig. 3, as a restrictive element to impose a pressure distribution and control the production along the entire length of the wellbore.

The screen liner was run in an optimum placement of the main sand from gas above and water below (Fig. 1), by use of a 3D simulation model (see **Fig. 4**). It is very crucial to mitigate coning tendency because both gas and water that prolong the life of the well are present in proximity to the well. This unique horizontal-well completion is different from the standard field completion, which calls for a cemented liner, and is selectively perforated.

Environmentally acceptable drilling and completion fluids have been developed and used to drill the horizontal hole section at the point at which it is mainly unconsolidated sandstone with sensitive shale stringers. The drilling- and completion-fluid formulations used will also be presented in this paper.

Case Study

Discussion. Spinner logs were run in three wells (of which Z-253 is one) in an effort to determine the completion efficiencies of each well type and to help optimize future horizontal-well completions. The production spinners were used to validate the effective producing length (L_{eff}) from each well. Quantifying L_{eff} helped reduce the error rate in estimating the mechanical skin damage from production-modeling simulation work. In addition, returning to permeability testing in the laboratory helped determine the permeability of the mud-filtrate-invaded zone. Using these data, along with resistivity data, to measure the depth of invasion from the mud filtrates, skin estimates were made by use of an inflow simulator.

Well Description. Two Z-253 neighboring wells were drilled and conventionally completed overbalanced as 7-in.-cemented-liner completions, through which the best-quality pay encountered was perforated by use of 3⅜-in. TCP guns. Both wells were perforated overbalanced in vicosified completion brine. A total of 700 ft of interval was perforated at 4 shots per foot in each well.

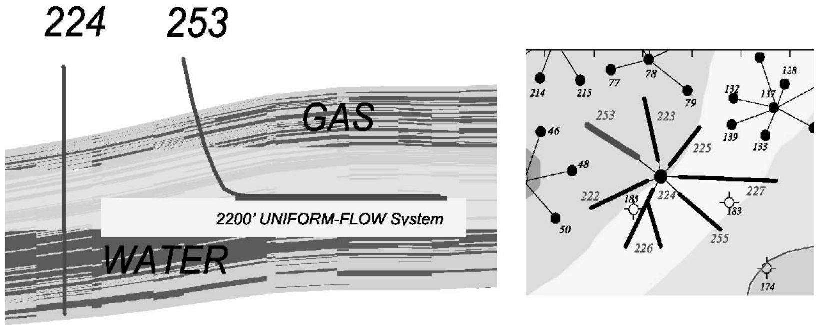

Fig. 1—Z-253 water/gas-contact optimum well placement, Z-253 platform map.

Z-253 was drilled overbalanced to a total depth of 9,657 ft, with a total of 2,200 ft across the sand reservoir, and completed with a sand-control screen and four mechanical ECPs. These packers were run to enhance the effect of the regulator by creating an annular barrier between sections of various permeabilities.

Drilling and Completion Fluids. The previous drilling problems encountered in the Z-field were studied and reviewed to design the most suitable and cost-effective drilling fluid to use in the Z-253 horizontal section. Moreover, reservoir-rock fluid characteristics were laboratory tested to determine the amount of damage caused by different drilling fluids and to select the most suitable fluid to be used in this field. An oil-based mud was selected and developed in-house to overcome troublesome holes while drilling shale and sand sections in Saudi Arabia's offshore field.

The high initial maintenance cost of using the invert emulsion mud is justified by the fact that drilling continues problem-free and the mud is recycled. Moreover, invert-emulsion-oil mud (IEOM) has an excellent stability and solids tolerance. The mud will provide a gauged hole while drilling, hole lubrication, and lower permeability impairment.

The specialized-field technical support services of Saudi Aramco's R&D center were used to monitor the drill-in fluid for the entire 2,200 ft of 8½-in. horizontal section into the sand reservoir. Solids-free emulsion/completion fluid was displaced into the hole before running the screen. The horizontal lateral has been successfully drilled and completed as planned—problem-free to the setting depth of 9,654 ft.

IEOM System. The target formation is mainly unconsolidated, highly permeable and porous sandstone, with water-sensitive shale. Shale swelling and inadequate mud density can lead to wellbore-stability problems.

IEOM is water in oil, with oil as a continuous phase. Invert emulsion has proved to be the right choice to drill the horizontal hole in the offshore Z-field. The base oil used in the formulation is locally produced, low in toxicity, and is an aromatic-content oil, which has a lethal-concentration-50% value of approximately 180,700 ppm, which meets U.S. Environmental Protection Agency guidelines for nontoxic drilling fluids.[1] The drill-in formulation per barrel for average properties is shown in order of addition in **Table 1.**

Additives used in formulating the invert emulsion were emulsifiers, organophilic clays as a viscosifier, and organophilic colloid as a filtrate-control additive.

Specialized-field technical services were used to closely monitor tests on the rigsite and recommend corrective chemical treatment of the mud on a timely basis, including fluid-loss readings, report analysis, calcium chloride concentration, rheological properties, and electrical stability (ES).

Solids-Free Oil-Mud (SFOM) Emulsion/Completion-Fluid Formulation. Solids-free emulsion/completion fluid was displaced into the hole before running the screen. The major components of the SFOM are low-toxicity oil, brine ($CaCl_2/CaBr_2$), and the emulsifier. With higher density required, the oil/brine ratio was adjusted to within 55/45 to 60/40 for maximum ES (ES of 400 V is minimum). To achieve the required weight, $CaBr_2$ was used in combination with $CaCl_2$. Completion-fluid formulation per barrel with the average properties is shown in order of addition in **Table 2.**

Production and Flowmeter Data

The first horizontal coiled-tubing-logging (CTL)/production-logging tool (PLT) logging job was successfully completed on this well. The preliminary analysis of the log indicated uniform-flow contribution across the horizontal section with total rates of ±8.0 and ±10.0 thousand RB/D at choke settings of 72/64 and 90/64 in., respectively. The estimated productivity index (PI) was approximately 160 STB/D/psi for the uniform-inflow-control-system completion and estimated to be 200 STB/D/psi for an openhole completion. It should be noted that two PIs are provided because there is a designed pressure drop built into the uniform-inflow-control system. Providing the second PI allows for the uniform-inflow-control-system completions to be compared to offset wells from a formation-damage viewpoint.

Simulator-Results Validation

The following simulation was run on the basis of the actual CTL/PLT pressure and rate data (preliminary evaluation): flowing-borehole pressure at 10,000 STB/D = 2,498 psi and shut-in borehole pressure = 2,557 psi.

The production distribution was obtained from the PLT log. The simulator results matched the PLT output, using permeability and skin as variables. The horizontal lateral is divided into four sections, and for each section, production contribution was matched.

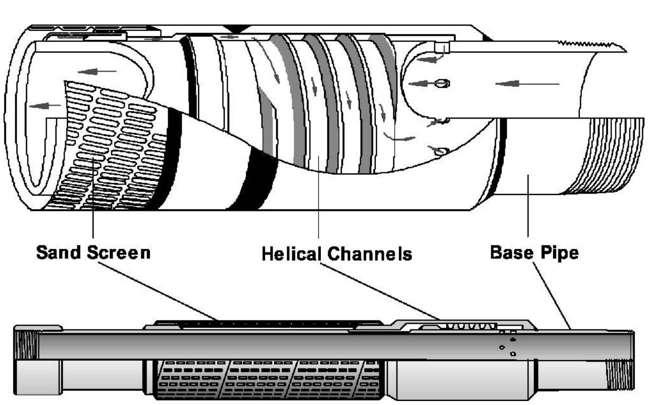

Fig. 2—Uniform-inflow-control system with detailed view of regulator section.

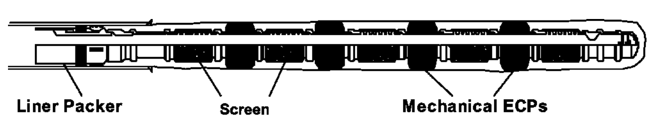

Fig. 3—Uniform-inflow-control system as run in Z-253.

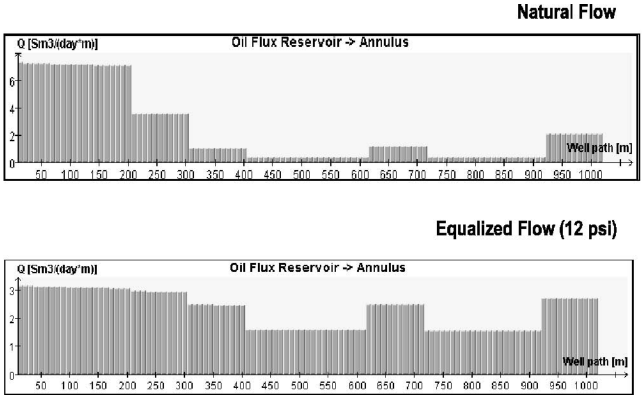

Fig. 4—Simulation results of uniform-inflow-control system in Z-253.

The uniform-inflow-control system installed has eliminated/minimized annular flow and ensured uniform inflow along the lateral at the cost of a minute pressure drop. The graph in **Fig. 5** illustrates the pressure profile at the sandface, which indicates the pressure drawdown applied to the formation and the pressure profile inside the system along the all-horizontal section (Line A). A cumulative PI at the sandface was calculated on the basis of the PLT data acquired, where P = PI:

$$P_{\text{sandface}} = P_1 + P_2 + P_3 + P_4, \quad \text{(1)}$$

and where $P_x = Q_x / D_x$

$$= 3{,}000/27+1{,}300//29+2{,}7000/5+3{,}000/9$$

$$= 1030 \text{ STB/D/psi.}$$

Fig. 6 plots the actual case with the uniform-inflow-control system (Line 1) vs. a conventional horizontal completion (Line 2). The uniform-inflow-control system allows the first 1,000 ft to contribute to the overall production.

Fig. 7 compares the actual well vs. a conventional completion, in which the first half would be contributing with only 10% of the overall production (Line 2) compared to 43% in the uniform-inflow-control-system case (Line 1).

Program Results

The results from the uniform-inflow-control-system completion are positive because the well has a higher PI and better inflow profile compared to offset-cemented and perforated completions. All three wells have been considered an economic success and will recover the targeted oil reserves. Z-253 being completed with a uniform-inflow-control system has the best completion efficiency, which has also been attributed to the proper well placement, drilling- and completion-fluids design, and the planned wellbore cleanup.

TABLE 2—SFOM COMPOSITIONS AND PROPERTIES

Formulation Description	Unit	Concentration
Low-toxicity oil	bbl	As needed
Primary emulsifier	lbf	1.5
Lime	lbf	4–6
Water	bbl	As needed
Secondary emulsifier	lbf	1–2
$CaBr_2$ (53%)	bbl	As needed
$CaCl_2$ (78%)	lbf	As needed
Density	lbm/ft^3	73–79
Yield point	lbf/100 ft^2	15–18
Oil/water ratio		55/45–60/40
Electrical stability	V	>400
Cl^-	ppm×10^3	350

TABLE 1—IEOM COMPOSITIONS AND PROPERTIES

Formulation Description	Unit	Concentration
Low-toxicity oil	bbl	As needed
Primary emulsifier	lbf	1.5–2
Lime	lbf	6–8
Organophilic colloid	lbf	6–8
Water	bbl	As needed
Organophilic clay	lbf	8–10
Secondary emulsifier	lbf	0.5–1
$CaCl_2$ (78%)	lbf	As needed
$CaCo_3$ "fine"	lbf	36–40
$CaCo_3$ "medium"	lbf	12–16
Density	lbm/ft^3	73–79
Plastic viscosity	cp	15–20
Yield point	lbf/100ft^2	20–25
10-second gel	lbf/100ft^2	4–8
10-minute gel	lbf/100ft^2	4–12
Filtrate high pressure/high temperature (250°F/500 psi)	mL/30min	1–2 (all oil)
Electrical stability	V	> 600
Chlorides	ppm×10^3	300–350
Oil/water ratio		70/30
Excess lime	lbm/bbl	4–6

Conclusions

1. The uniform-inflow-control system proved to be a cost-effective and economically feasible completion, saving time and money compared with conventional perforated casing-completion techniques.
2. By eliminating perforation guns on the rig, the uniform-inflow-control-system completion creates a safer operation compared with perforated-casing completion.
3. The results from the uniform-inflow-control-system completion are positive because the well has a higher PI and better inflow profile compared with neighboring cemented and perforated conventional completions.
4. The simulator results have matched the CTL/PLT output by use of permeability and skin as variables.

Nomenclature

D = pressure drop, psi
L_{eff} = effective producing length from net pay within the horizontal section
P_{sandface} = cumulative PI at the sandface
P_1 = Q_1/D_1
P_2 = Q_2/D_2
P_3 = Q_3/D_3
P_4 = Q_4/D_4
Q = flow rate, STB/D

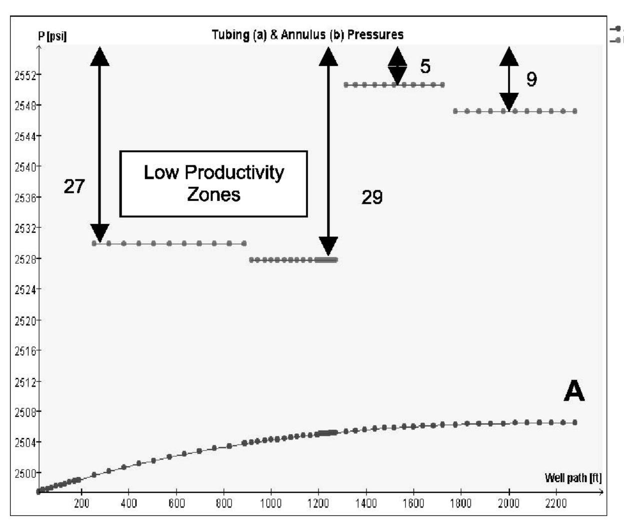

Fig. 5—Tubing and annulus pressures.

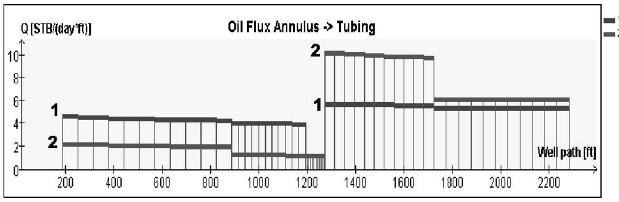

Fig. 6—Oil-flux annulus→tubing.

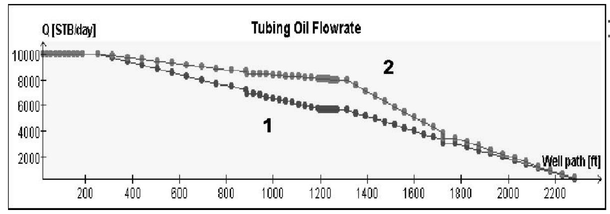

Fig. 7—Tubing oil flow rate.

Acknowledgments

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References

1. Ayers, R.C. Jr, Sauer, T.C. Jr., and Anderson, P.W.: "The Generic Mud Concept for NPDES Permitting of Offshore Drilling Discharge," *JPT* (March 1985) 475.

SI Metric Conversion Factors

bbl ×	1.589 873	E–01	=	m^3
cp ×	1.0*	E–03	=	Pa·s
ft ×	3.048*	E–01	=	m
°F	(°F–32)/1.8		=	°C
in. ×	2.54*	E+00	=	cm
lbf ×	4.448 222	E+00	=	N
lbm/bbl ×	2.853 010	E+00	=	kg/m^3
lbm/ft^3 ×	1.601 846	E+01	=	kg/m^3
lbf/100 ft^2 ×	4.788 026	E–01	=	Pa
m^3/d ×	1.589 873	E–01	=	m^3/s
mL ×	1.0*	E+00	=	cm^3
psi ×	6.894 757	E+00	=	kPa
STB/D/psi ×	1.589 873	E–01	=	STM^3

*Conversion factor is exact.

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